

1 AVAW_CV-Waste: An Offline Computer Vision 2 Annotation Tool for Waste Management Research

3 **Gerald Koinig^{1¶} and Bojan Lorber¹**

4 ¹ Montanuniversität Leoben, Austria ¶ Corresponding author

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5 Summary

6 AVAW_CV-Waste is a free and open-source offline desktop application for creating and correcting
7 image annotations used to train object detection models. The software allows users to draw,
8 edit, move, resize, copy, and manage bounding boxes around objects in images. It creates
9 outputs compatible with YOLO-based artificial intelligence workflows, specifically the YOLO
10 PyTorch TXT annotation format commonly used by YOLO11, YOLOv8, YOLOv5, and related
11 computer vision models.

12 CV-Waste supports the integration of existing YOLO models for semi-automatic annotation
13 generation, reducing the amount of manual labelling required during dataset preparation. This
14 assistance can be used to annotate complete images or follow a one-click annotation process in
15 which the user clicks on an object and the currently loaded model tries to fit a bounding box
16 around the selected object. The application was designed specifically for waste management,
17 recycling, and material sorting research, where datasets often contain highly domain-specific
18 object classes and cannot easily be processed using generic or cloud-based annotation platforms.

19 Because the software operates entirely offline, it can be deployed directly at industrial facilities or
20 research sites where internet access may be limited or unavailable. Local processing additionally
21 guarantees data sovereignty, an important requirement for industrial research collaborations
22 involving sensitive operational data.

23 The primary goal of AVAW_CV-Waste is to simplify and accelerate dataset creation for non-
24 specialist users, including students, laboratory staff, recycling operators, and industrial research
25 partners. By reducing the technical complexity of annotation workflows, the software supports
26 the development of custom machine learning models for applications such as waste classification,
27 contaminant detection, scrap sorting, and sensor-based recycling research.

28 Statement of need

29 CV-Waste was developed as a free and open-source alternative that can be adapted to workflows
30 in waste management and recycling research.

31 Currently cloud-based commercial services exist and provide server-side automated annotation
32 using existing pre-trained models. However, such functionality is often confined to paid
33 subscription fees and restricted to the models supported by the respective platform.
34 Consequently, researchers are frequently unable to leverage their own waste management-
35 specific pre-trained models to support and aid annotation. Further, as many existing tools are
36 cloud-based, data sovereignty may not always be guaranteed. This is an increasingly precarious
37 issue when working with industrial partners in general and partners in waste management in
38 particular. These data sovereignty issues may prohibit the use of these commercial, cloud-based
39 alternatives outright. CV-Waste allows users to define their own custom annotation classes,
40 load and use their own models for automatic predictions and to work entirely offline without

41 requiring cloud-services.
42 Open source alternatives include frameworks for building annotation tools (Epifânio et al.,
43 2022). However, CV-Waste addresses a different practical need. Its purpose is not to be a
44 programmable annotation framework, but to provide an immediately usable, domain-focused
45 desktop GUI for YOLO-style waste annotation workflows in waste management research.
46 CV-Waste is designed for research assistants, students, laboratory staff, and industrial partners
47 who may not have programming experience, but still need to create, review, and correct
48 high-quality annotations efficiently. In this sense, (Epifânio et al., 2022) serves developers
49 and users who want to build or customize annotation tools, whereas AVAW_CV-Waste
50 serves non-programming users who need a polished, accessible, and research-ready annotation
51 environment. AddaxAI is primarily a no-code platform for training and deploying YOLOv5
52 object detection models (Lunteren, 2023). CV-Waste focuses specifically on creating, reviewing,
53 and correcting YOLO-style bounding-box annotations for waste and recycling datasets, which
54 can then be used to train most object classification networks including YOLO26 and RT-DETR
55 models.

56 SAMBA targets semantic segmentation rather than bounding-box object detection, while LaMa
57 and occupationMeasurement are aimed at labelling or coding text-based qualitative and survey
58 data (Bogachenkova et al., 2023; Docherty, 2024; Simson et al., 2023). Therefore, CV-
59 Waste provides a dedicated, lightweight, offline, and research-oriented GUI for bounding-box
60 annotation in waste management applications and fills a distinct gap.

61 State of the field

62 The field of leveraging object classification models like YOLO or RT-DETR for classifying waste
63 particles is a recently emerging one. Low cost sensoric equipment, i.e. RGB cameras, have
64 becoming of increasing interest with the advent of attainable post-consumer hardware capable
65 of running these models in deployment scenarios. The systems are becoming increasingly
66 adopted to pre-sort waste streams prior to more chemically astute systems like LIBS or XRF,
67 XRT. However, as the training of waste classification models requires extensive, diverse and well
68 labelled datasets that represent the expected working environment i.e. deformed, heterogeneous,
69 contaminated, particles on a conveyor belt, existing datasets often cannot be used to train these
70 models. Albeit, object classification models are becoming increasingly adopted to sort highly
71 specific object classes, such as plastics, paper, scrap, hazardous waste, or mixed waste fractions.
72 Sorting tasks that were recently primarily performed manually. This reliance on manual sorting
73 of waste fractions is another driving factor in the adoption of object classification models
74 in waste management, as finding and holding of staff is becoming increasingly difficult for
75 MRF operators. Additionally, scaling up manual sorting operations to handle the increasing
76 volume of waste streams is unfeasible. Consequently, large stakeholders are already investing
77 in research projects regarding the application of object detectors for waste sorting. This
78 includes lighthouse projects like StrateX and Scarpa which focus on Textiles and Footwear
79 waste, KIRAMET, which researches scrap recovery for the carbon neutral transformation of steel
80 production and greenPLAST-food, which aims at improving the circular economy of lightweight
81 packaging waste, as it is currently polluting environment and wildlife. All these projects need
82 extensive datasets to train their models and are using CV-Waste to create them.

83 Software design

84 While generative AI can now accelerate the initial development of software like this, extensive
85 testing in real research environments is still required. In particular, evaluating the tool with
86 researchers of different technical backgrounds cannot be meaningfully outsourced or offloaded
87 to generative AI systems.

88 Thus, the UI and UX of this tool were subject to constant iteration and improvement, spanning

multiple years of research projects in this emerging field in waste management research. It could therefore incorporate criticism and feedback from students during lectures, as well as from researchers and workers that annotated tens of thousands of images using this tool. Ongoing use and the evolving requirements of research projects, particularly in waste-management-related sorting tasks, will inevitably uncover further opportunities for optimisation and adaptation. The use of Python, the widely adopted UI framework Tkinter, and an object-oriented software architecture is intended to support straightforward customisation and extension by researchers in waste management and recycling domains.

To maintain simplicity and accessibility, the software intentionally avoids implementation complexity, such as multithreading or multitasking even though these approaches could have potentially improved application responsiveness.

Since Python is already widely used in machine learning and computer vision research, the tool integrates naturally into existing scientific workflows.

Several design trade-offs were made to ensure broad hardware compatibility and stable operation on lower-performance research machines while still providing the quality-of-life features expected from such a tool. Lightweight rendering and simplified interaction workflows were prioritised over computationally expensive visualisation features. Although the default Python environment is primarily designed for CPU-based inference, the underlying inference modules also enable advanced users to leverage available GPU hardware resources with only minor changes to the installation workflow.

Lastly, CV-Waste was specifically tailored to bounding box annotation. Bounding Box annotation is itself tailored to waste management applications as it is faster than segmentation-based annotation and requires less computational hardware when deployed. Additionally usual ejection mechanisms like high pressure nozzle bars or flippers do not have the spatial resolution to leverage the additional localisation detail provided by segmentation masks.

Research impact statement

AVAW_CV-Waste has already been successfully used in the development of scientific datasets and in research related to waste detection, classification, and recycling workflows.

AVAW_CV-Waste addresses a major bottleneck in the development of computer vision systems for waste management and recycling research, namely the creation of annotated training datasets.

AVAW_CV-Waste is currently being used within multiple recycling and circular economy lighthouse research projects, including KiRAMET, StraTex, greenPLAST-food, and Scarpa.

Within these projects, the software has been applied in research workflows involving post-consumer textiles, lightweight packaging waste, and post-shredder scrap, where large quantities of annotated image data are required for training and evaluating object detection models.

Datasets created using AVAW_CV-Waste contributed to research on sensor-based sorting of shredder scrap towards green steel production ("AI Pipeline for Sensor Based Sorting of Shredder Scrap Towards Green Steel Production," 2026), AI-assisted recycling workflows for metal composite waste streams (KiRAMET, 2026; Kiramet, 2026), and plastic recycling systems for food-contact materials (GreenPLAST-Food, 2026).

The generated datasets also supported research on robust YOLO-based ejection of copper-containing particles from heavily corroded scrap streams ("Robust YOLO-Based Ejection of Copper-Containing Particles in Heavily Corroded Scrap Towards Green-Steel Production," 2026), detection of copper-containing scrap in post-shredder fractions using machine vision and artificial intelligence (Detection of Copper-Containing Scrap in a Post-Shredder Fraction with Machine Vision and Artificial Intelligence Towards Green-Steel Production, 2026), and deep learning approaches for classification of copper-containing metal scrap in recycling processes

(*Deep Learning Approaches for Classification of Copper-Containing Metal Scrap in Recycling Processes*, 2026).

These applications demonstrate the utility of the software for reducing manual annotation effort and enabling machine learning methods for heterogeneous waste sorting tasks.

AI usage disclosure

Generative artificial intelligence tools were not used for the architectural design or core software implementation of AVAW_CV-Waste.

ChatGPT (OpenAI) was used during software development for documentation support, generation of docstrings, bug-fixing assistance, and inline code modifications. For this manuscript, generative AI tools were used only for grammatical improvements and formatting support, including conversion of manuscript content from .docx to Markdown syntax.

All generated content and code modifications were manually reviewed and validated by the authors.

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